Telecom Customer Churn & Revenue Optimization Analysis

1. Project Overview

This project analyses customer retention patterns for a telecommunications provider using a dataset of subscriber transactions. The primary objective is to identify critical churn drivers, quantify the financial impact of at-risk revenue, and provide data-driven recommendations to stabilize the monthly revenue stream.

2. Dataset Summary

- Total Customers Analysed: 7,043
- Key Metrics: Churn Rate %, At-Risk Revenue, Lost Monthly Revenue, and Tenure Groups.
- Customer Features: Demographics (Senior Citizen, Gender), Account Info (Contract Type, Payment Method, Paperless Billing), and Services (Internet Service, Tech Support, Device Protection).

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info () to check structure and df.describe () for summary statistics.

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup
count	7043	7043	7043.000000	7043	7043	7043.000000	7043	7043	7043	7043	7043
unique	7043	2	NaN	2	2	NaN	2	3	3	3	3
top	7590-VHVEG	Male	NaN	No	No	NaN	Yes	No	Fiber optic	No	No
freq	1	3555	NaN	3641	4933	NaN	6361	3390	3096	3498	3088
mean	NaN	NaN	0.162147	NaN	NaN	32.371149	NaN	NaN	NaN	NaN	NaN
std	NaN	NaN	0.368612	NaN	NaN	24.559481	NaN	NaN	NaN	NaN	NaN
min	NaN	NaN	0.000000	NaN	NaN	0.000000	NaN	NaN	NaN	NaN	NaN
25%	NaN	NaN	0.000000	NaN	NaN	9.000000	NaN	NaN	NaN	NaN	NaN
50%	NaN	NaN	0.000000	NaN	NaN	29.000000	NaN	NaN	NaN	NaN	NaN
75%	NaN	NaN	0.000000	NaN	NaN	55.000000	NaN	NaN	NaN	NaN	NaN
max	NaN	NaN	1.000000	NaN	NaN	72.000000	NaN	NaN	NaN	NaN	NaN

DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
7043	7043	7043	7043	7043	7043	7043	7043.000000	7043	7043
3	3	3	3	3	2	4	NaN	6531	2
No	No	No	No	Month-to-month	Yes	Electronic check	NaN	20.2	No
3095	3473	2810	2785	3875	4171	2365	NaN	11	5174
NaN	NaN	NaN	NaN	NaN	NaN	NaN	64.761692	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	30.090047	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	18.250000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	35.500000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	70.350000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	89.850000	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	118.750000	NaN	NaN

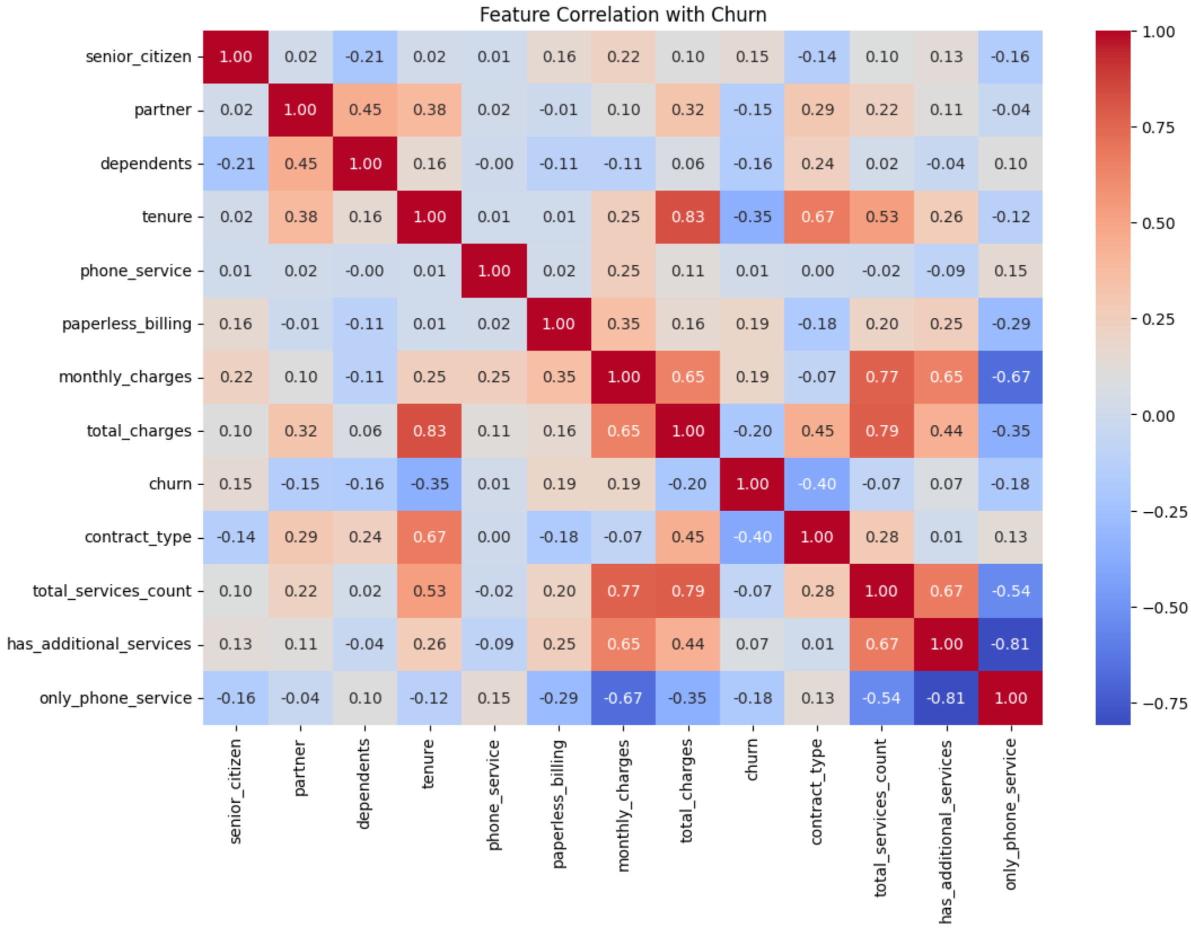
- **Typecasting:** Checked data type of strings and numeric and converted TotalCharges column to numeric datatype.

- **Missing Data Handling:** Checked for null values and imputed missing values in the TotalCharges column with 0 as tenure was 0.

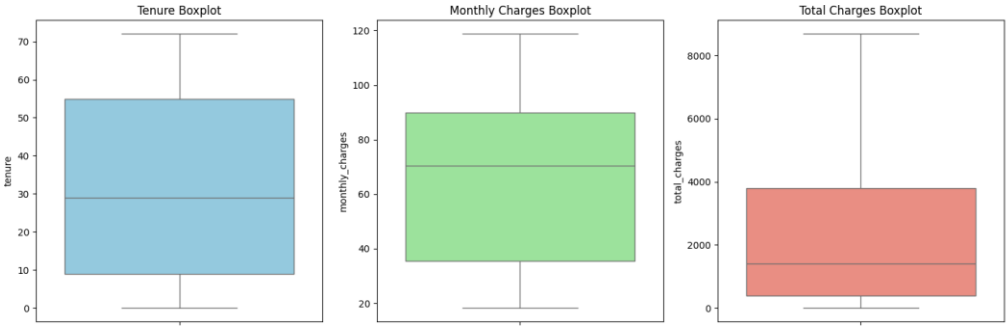
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:**
  - Created numeric columns from text data.
  - Created **tenure_group** column by binning tenure.
  - Created **total_services_count, has_additional_services** columns for analysis.

- **Correlation:** Checked the correlation between churn and numeric values.



- **Outliers:** Checked whether the data has any outliers or not with the help of box plot.



- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in MySQL to answer key business questions:

1. **Churned Customers and Lost Revenue by Contract Type –** Calculated which contract type is responsible for the highest number of churned customers and the greatest monthly recurring revenue loss.

contract_type	churned_customers	lost_mrr
0	1655	120847
2	48	4165
1	166	14118

2. **Overall Churn Rate -** Calculated percentage of all churned customers.

churn_rate
26.54

3. **Churn by Senior Citizen Status -** Calculated whether senior citizens are more likely to churn than non-senior customers.

senior_citizen	count	churn
1	16.21	41.68
0	83.79	23.61

4. **Churn by Tenure Group -** Calculated the stage of the customer lifecycle is churn most likely to occur.

tenure_group	count	churn
New	31.04	47.44
Reliable	14.54	28.71
Established	22.63	20.39
Loyal	31.79	9.51

5. **Churn by Phone Service and Internet Service Combination -** Calculated combinations of phone and internet services with the highest churn rates.

phone_service	internet_service	count	churn
1	Fiber optic	43.96	41.89
0	DSL	9.68	24.93
1	DSL	24.69	16.62
1	No	21.67	7.40

6. **Churn by Internet Service Type -** Calculated which internet service type (DSL, Fiber Optic, No Internet) has the highest churn rate.

internet_service	count	churn
Fiber optic	43.96	41.89
DSL	34.37	18.96
No	21.67	7.40

7. **Churn by Total Number of Services -** Calculated how does the number of subscribed services impact churn.

total_services_count	count	churn
2	13.59	37.72
1	16.44	34.46
3	13.89	31.29
4	13.25	25.72
5	10.25	21.75
0	23.67	20.76
6	5.96	11.67
7	2.95	5.29

8. **Additional Services Indicator -** Calculated whether customers with value-added services (such as security, backup, or support) have lower churn rates or not.

has_additional_services	count	churn
1	76.33	28.33
0	23.67	20.76

9. **Churn by Contract Type -** Calculated how does contract duration influence customer churn.

contract	count	churn
Month-to-month	55.02	42.71
One year	20.91	11.27
Two year	24.07	2.83

10. **Churn by Payment Method -** Calculated which payment methods are associated with the highest churn rates.

payment_method	count	churn
Electronic check	33.58	45.29
Mailed check	22.89	19.11
Bank transfer (automatic)	21.92	16.71
Credit card (automatic)	21.61	15.24

11. **Churn by Paperless Billing -** Calculated whether paperless billing correlates with higher or lower churn.

paperless_billing	count	churn
1	59.22	33.57
0	40.78	16.33

12. **Streaming Segment Analysis -** Calculated which customer segment (Movies only, TV only, Both, or None) has the highest churn rate and average monthly charges?

streaming_segment	count	churn	avg_monthly_charges
Tv	10.89	31.68	7735.28
Movies	11.25	31.19	7681.17
Both	27.55	29.43	9323.75
None	50.32	22.80	4375.6

13. **Internet Service + Streaming Segment -** Calculated within each internet service type, how does streaming behaviour influence churn among month-to-month customers.

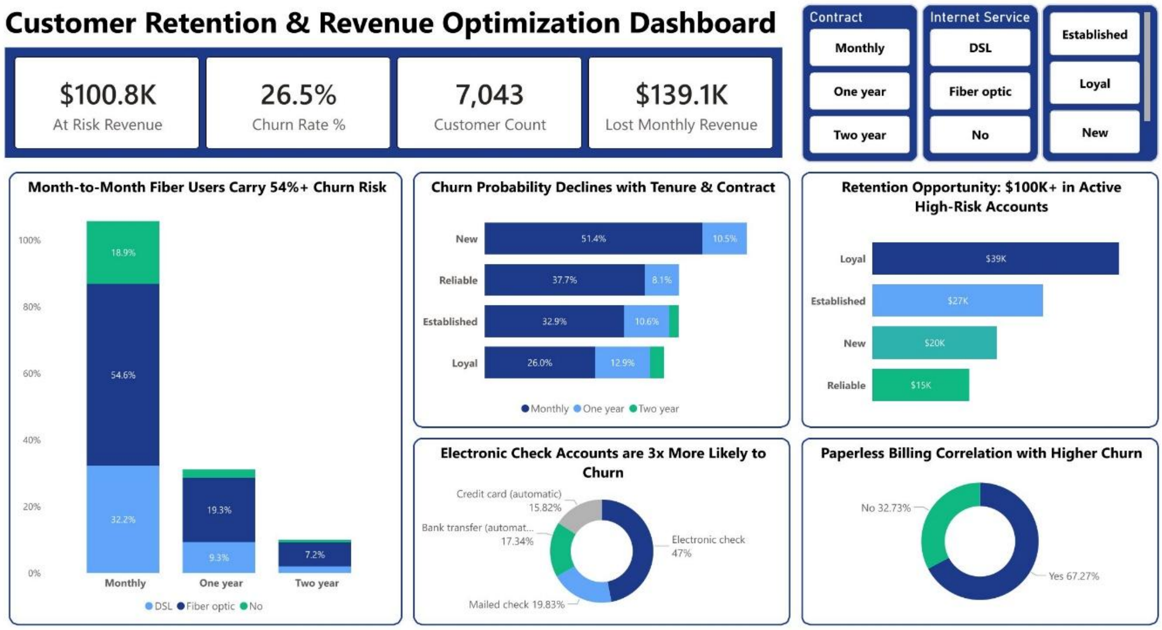
internet_service	streaming_segment	count	churn	avg_monthly_charges
Fiber optic	Both	9.40	57.70	9963.04
Fiber optic	Tv	4.80	55.03	8767.4
Fiber optic	Movies	4.76	53.13	8749.64
Fiber optic	None	11.26	52.46	7601.6
DSL	None	10.92	33.42	4559.6
DSL	Both	2.14	31.79	6207.12
DSL	Movies	2.21	30.77	5541.22
DSL	Tv	2.09	27.89	5672.18
No	None	7.44	18.89	2040.95

Structured analysis in Power BI answered key business questions regarding churn drivers:

1. Contractual Risk: Month-to-Month contracts exhibit the highest churn probability at 51.35% for new customers.
2. Service Vulnerability: Fiber Optic subscribers on Month-to-Month plans carry a massive 54.6% churn risk.
3. Financial Impact: There is currently $100.8K in active revenue categorized as "At Risk," primarily concentrated in the "Loyal" and "Established" tenure groups.
4. Payment Friction: Accounts using Electronic Checks are 3x more likely to churn (47%) compared to automated payment methods.
5. Retention Opportunity: Churn rates decline significantly with tenure; "Loyal" customers (72+ months) show only a 26.02% churn rate on Month-to-Month plans.

5. Power BI Dashboard

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Strategic Business Recommendations

Based on the data findings, the following actions are recommended to optimize revenue:

- **Incentivize Contract Migrations:** Targeted promotions to move Month-to-Month Fiber users into 1-year contracts to halve churn risk.
- **Address Payment Friction:** Offer minor incentives for "Electronic Check" users to switch to automated "Bank Transfer" or "Credit Card" billing.
- **Bundle Tech Support:** Proactively offer Tech Support services to new Fiber Optic customers, as data suggests this significantly increases "stickiness."
- **Retention Marketing:** Focus loyalty rewards on "Established" and "Reliable" groups to protect the $100K+ in active at-risk revenue.